

Restaurant Performance Analysis

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Tools Used:

- Power BI
- CSV Datasets: food, menu, orders, restaurant

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Introduction

The purpose of this project was to analyze restaurant sales performance, customer ratings, cuisine popularity, and revenue trends using Power BI. Using multiple restaurant-related datasets, the analysis aimed to identify high performing restaurants, understand customer preferences, evaluate geographic sales patterns, and provide data driven business recommendations. The interactive dashboard was designed to support decision-making by highlighting key performance metrics, operational trends, and customer behavior insights.

Research Questions

- Which restaurants generate the highest revenue?
- Which cities have the highest sales volume?
- What cuisines are ordered the most frequently?
- Which restaurants have the highest customer ratings?
- Are higher-rated restaurants associated with higher sales?
- Which restaurants have the highest average order value?

Hypotheses

1. Restaurants with higher ratings generate more revenue.
2. Fast food and beverage restaurants receive higher order volume.
3. Restaurants located in larger cities outperform smaller city restaurants.

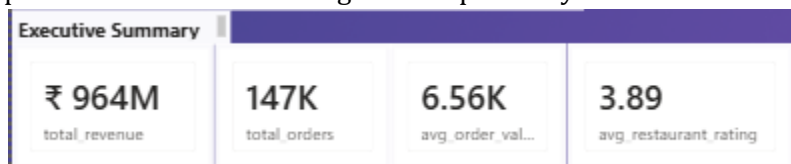
Data Cleaning Strategy

- Remove rows with missing restaurant IDs because they cannot be connected to restaurant performance metrics.

- Remove negative and zero value sales transactions to improve revenue accuracy.
- Remove rows with missing restaurant names or cuisine information to ensure complete categorization and visualization.
- Validate date formatting to ensure compatibility with Power BI time series analysis.
- Check for duplicates prior to analysis; no major duplicate issues were identified.
- Replace invalid rating values such as "--" with null values so the ratings column could be converted to a numeric data type.
- Convert rating fields to Decimal Number format to support average rating calculations and correlation analysis.
- Convert sales and revenue fields to Fixed Decimal Number format for accurate aggregations and currency formatting.
- Identify and handle data conversion errors caused by text values or improperly formatted numeric fields.

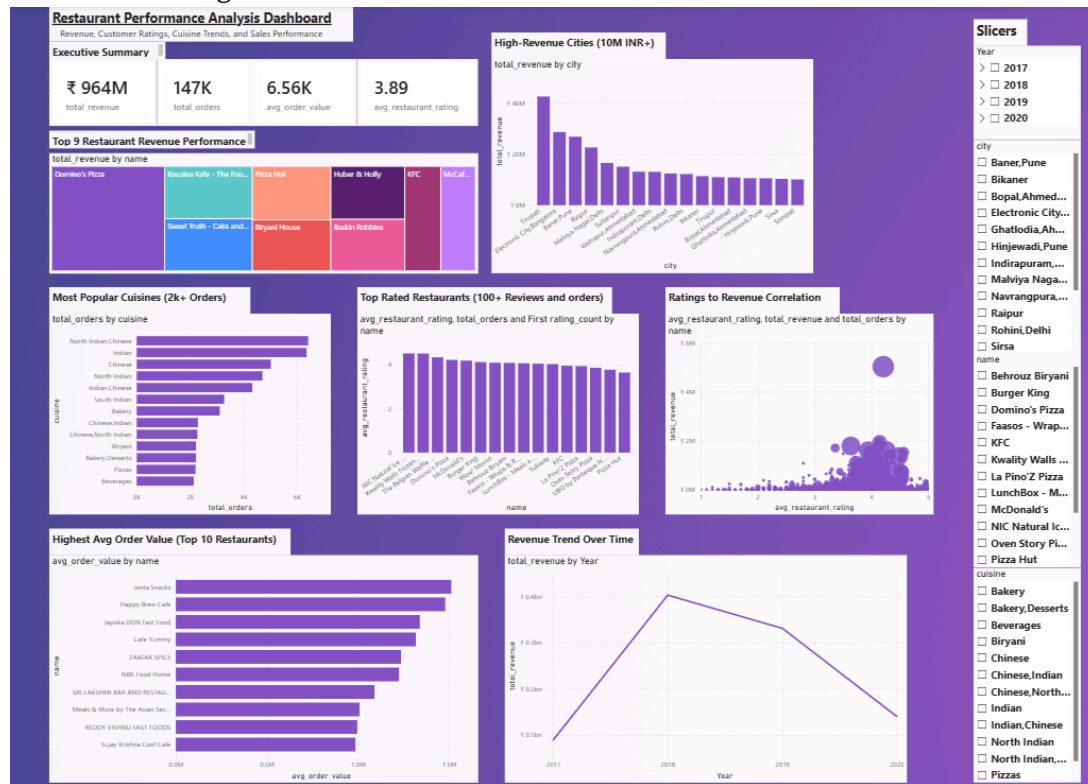
Dashboard Overview

The Power BI dashboard was designed to provide an interactive overview of restaurant business performance across revenue, customer ratings, cuisine popularity, and geographic sales trends. The dashboard begins with an Executive Summary section containing high level KPI cards for total revenue, total orders, average order value, and average restaurant rating. This section provides a quick overview of overall business performance before moving into deeper analysis sections.



The remaining dashboard sections focus on restaurant revenue performance, geographic sales analysis, cuisine popularity, customer ratings, revenue correlation analysis, and revenue trends over time. Interactive slicers were added to improve dashboard usability and allow users to dynamically filter visuals by year, city, cuisine, and restaurant name. These slicers enable deeper exploration of the aforementioned analysis areas and can be

seen on the far right of the dashboard.



Key metrics and calculated fields used throughout the dashboard include:

- Total Revenue = $\text{SUM}(\text{sales_amount})$
- Total Orders = $\text{COUNTROWS}(\text{orders})$
- Total Quantity Sold = $\text{SUM}(\text{sales_qty})$
- Average Order Value = $\text{Total Revenue} \div \text{Total Orders}$
- Average Restaurant Rating = $\text{AVERAGE}(\text{rating})$
- Revenue by Restaurant = Total Revenue grouped by restaurant name
- Revenue by City = Total Revenue grouped by city
- Revenue by Cuisine = Total Revenue grouped by cuisine
- Cuisine Popularity = Total Quantity Sold grouped by cuisine
- Top Restaurants by Revenue = Top N restaurants ranked by Total Revenue
- High-Revenue Cities = Cities with Total Revenue above 10M
- Ratings vs Revenue = Comparison of Average Rating against Total Revenue

Key Findings

Research Question 1

Which restaurants generate the highest revenue?



Visualization:

Discussion:

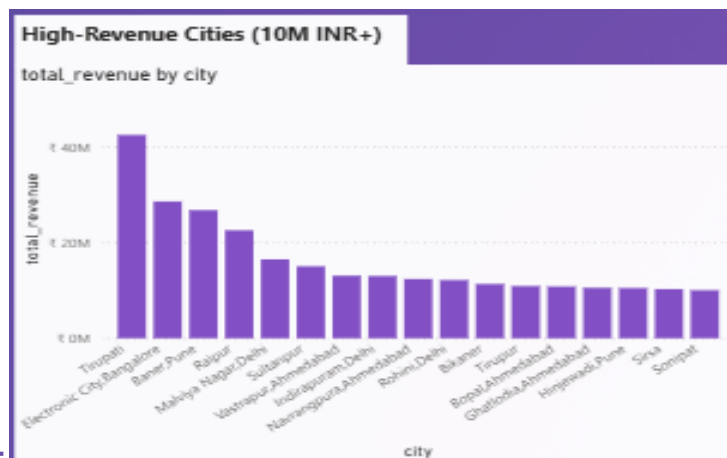
The restaurant revenue analysis showed that 7 out of the top 9 highest-performing restaurants were major Western fast food chains. Domino's Pizza generated the highest overall revenue among all restaurants included in the dataset. The strong performance of large chain restaurants suggests that brand recognition, widespread availability, and established customer trust may significantly influence revenue generation.

Answer to Research Question:

Domino's Pizza generated the highest overall revenue, and Western fast-food chains dominated the top performing restaurants in the dataset.

Research Question 2

Which cities have the highest sales volume?



Visualization:

Discussion:

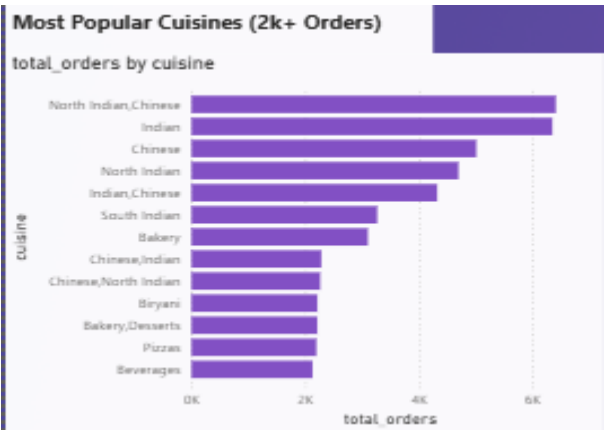
Tirupathi, Electronic City Bangalore, Baner Pune, and Raipur all generated more than ₹20 million INR in revenue, making them the strongest-performing cities in the dataset. These findings suggest that certain urban areas contribute disproportionately to overall restaurant sales and may represent regions with especially strong customer demand and ordering activity.

Answer to Research Question:

Tirupati, Electronic City Bangalore, Baner Pune, and Raipur generated the highest sales volume among the cities analyzed.

Research Question 3

What cuisines are ordered the most frequently?



Visualization:

Discussion:

The cuisine analysis demonstrated that customer demand is distributed across both traditional cuisine categories and globally recognized fast-food products, highlighting the importance of menu diversity and convenience within the restaurant industry. Beverage and baked goods categories also demonstrated high ordering frequency, suggesting strong complementary purchasing behavior alongside meals.

Answer to Research Question:

North Indian, Chinese, South Indian, Biryani, Baked Goods, Pizza, and Beverage categories were the most frequently ordered cuisines with at least 2000 orders.

Research Question 4

Which restaurants have the highest customer ratings?



Visualization:

Discussion:

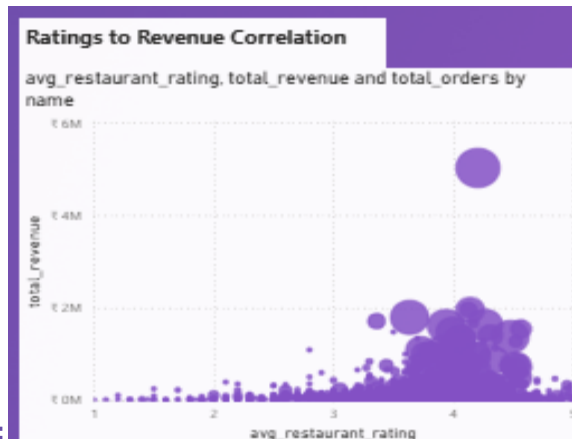
The customer ratings analysis revealed that several restaurant brands were able to maintain ratings above 4.0 while also supporting a high volume of customer orders and reviews. To improve reliability and avoid misleading comparisons from restaurants with very few reviews, the visualization was filtered to include only restaurants with more than 100 orders and customer ratings. One notable pattern in the findings was the strong presence of major fast-food chains and dessert-focused brands among the highest-rated restaurants. This may suggest that standardized menus, consistent preparation processes, recognizable branding, and efficient delivery operations contribute positively to customer satisfaction.

Answer to Research Question:

Several restaurants maintained customer ratings above 4.0 while also exceeding 100 orders and reviews, including Domino's Pizza, McDonald's, Burger King, Subway, and multiple dessert-focused brands. These restaurants demonstrated strong customer satisfaction alongside high customer engagement and order volume.

Research Question 5

Are higher-rated restaurants associated with higher sales?



Visualization:

Discussion:

The scatterplot analysis suggested a positive correlation between higher customer ratings and higher restaurant revenue. Restaurants with stronger ratings generally appeared to generate greater revenue, indicating that customer satisfaction may positively influence sales performance and repeat business behavior.

Answer to Research Question:

The analysis suggested a positive correlation between higher restaurant ratings and increased revenue performance.

Research Question 6

Which restaurants have the highest average order value?



Visualization:

Discussion:

The top 10 restaurants with the highest average order values each had only one recorded order and displayed values above ₹984K INR. These values appeared to be

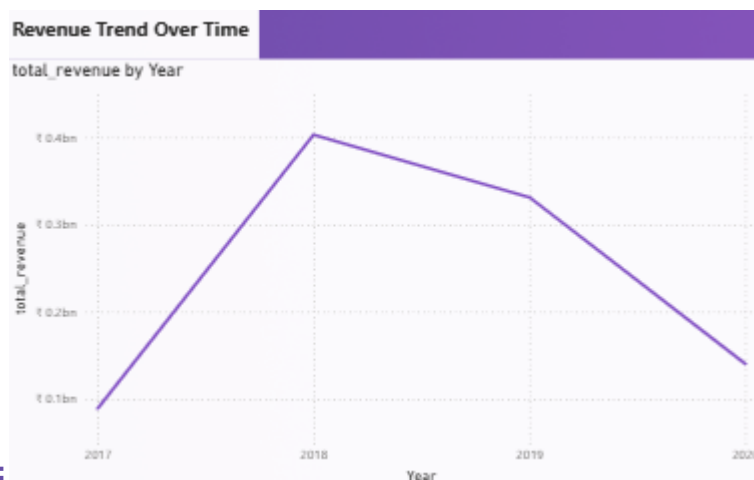
statistical outliers and did not align with the broader restaurant performance trends shown throughout the dashboard. Due to the extremely low order counts and unusually large transaction values, these findings were not considered reliable indicators of typical customer spending behavior.

Answer to Research Question:

Although several restaurants showed extremely high average order values, the results were determined to be outliers caused by unusually large single order transactions.

Research Question 7

How did revenue trends change over time?



Visualization:

Discussion:

The revenue trend analysis showed a significant increase in restaurant revenue between 2017 and 2018, growing from approximately ₹89M INR to ₹400M INR. Revenue then steadily declined to approximately ₹330M INR in 2019 and ₹140M INR in 2020. The decrease observed after 2018 may reflect changing market conditions and possible impacts from the COVID-19 pandemic on restaurant operations and customer purchasing behavior.

Answer to Research Question:

Restaurant revenue increased substantially between 2017 and 2018 before steadily declining through 2020.

Hypothesis Evaluation

Hypothesis 1

Restaurants with higher ratings generate more revenue.

Status: Accepted

Discussion:

The analysis supported the hypothesis that restaurants with higher customer ratings generally generate greater revenue. The scatterplot visualization showed a positive relationship between customer ratings and restaurant revenue, with many of the higher-rated restaurants also appearing among the strongest revenue performers.

Additionally, several restaurants with ratings above 4.0 and more than 100 orders and reviews consistently ranked among the top-performing businesses in the dataset. Restaurants such as Domino's Pizza, McDonald's, Burger King, and Subway demonstrated both strong customer satisfaction and strong sales performance. These findings suggest that customer satisfaction may positively influence customer retention, repeat purchases, and overall restaurant revenue.

Hypothesis 2

Fast food and beverage restaurants receive higher order volume.

Status: Accepted

Discussion:

The findings supported the hypothesis that fast-food and beverage-related restaurants receive higher order volume. The cuisine popularity analysis showed that Pizza and Beverage categories ranked among the most frequently ordered cuisines in the dataset. Additionally, major fast-food chains dominated the highest revenue-performing restaurants.

Restaurants such as Domino's Pizza, McDonald's, Burger King, and Subway consistently appeared among the strongest performers by both revenue and customer engagement. Beverage and dessert focused brands also showed strong customer demand, suggesting that convenience, accessibility, and recognizable branding contribute significantly to order volume within the restaurant industry.

Hypothesis 3

Restaurants located in larger cities outperform smaller-city restaurants.

Status: Accepted

Discussion:

The geographic sales analysis supported the hypothesis that restaurants located in larger or more commercially active cities outperform those in smaller cities. Cities such as Tirupati, Electronic City Bangalore, Baner Pune, and Raipur each generated more than ₹20 million INR in revenue and significantly outperformed many other cities included in the dataset.

These findings suggest that densely populated urban areas or highly active commercial regions contribute substantially to restaurant revenue performance due to higher customer demand, larger customer populations, and increased ordering activity. The results also indicate that geographic location plays an important role in overall restaurant success and revenue generation.

Conclusion

This project successfully analyzed restaurant performance using Microsoft Power BI by examining revenue trends, customer ratings, cuisine popularity, and geographic sales performance. The interactive dashboard provided a comprehensive view of restaurant business activity and helped identify meaningful patterns in customer preferences, restaurant success, and operational performance. The overall findings supported the project hypotheses. The analysis indicated that restaurants with higher customer ratings generally produced stronger revenue performance, suggesting that customer satisfaction may positively influence repeat business and long-term sales growth. Restaurants that maintained ratings above 4.0 while also supporting high order volume consistently appeared among the strongest performers in the dataset. The findings also supported the hypothesis that fast-food and beverage-related restaurants receive higher order volume. Major fast-food chains such as Domino's Pizza, McDonald's, Burger King, and Subway ranked among the highest-performing restaurants by both revenue and customer engagement. Beverage, dessert, and baked goods categories also demonstrated consistently strong order frequency throughout the analysis. Additionally, the geographic sales analysis suggested that restaurants located in larger or more commercially active cities tend to outperform those in smaller markets. Cities such as Tirupati, Bangalore, Pune, and Raipur generated exceptionally high revenue totals, highlighting the importance of geographic demand concentration and urban customer activity in restaurant sales performance. Revenue trend analysis further revealed substantial growth between 2017 and 2018 followed by a gradual decline through 2020. This decrease may reflect changing market conditions and possible external disruptions, including the operational and consumer impacts associated with the COVID-19 pandemic. Overall, the dashboard and supporting analysis demonstrate how business intelligence tools can transform raw restaurant data into actionable insights that support marketing strategies, operational planning, customer engagement efforts, and future business decision making.

Recommendations

Operational Recommendations

- Increase operational focus and staffing in high performing cities such as Tirupati, Bangalore, Pune, and Raipur, where restaurant revenue was significantly higher than other locations.
- Monitor high volume restaurants closely to maintain service quality, reduce wait times, and improve customer satisfaction during peak ordering periods.

- Continue tracking revenue trends over time to better prepare for seasonal fluctuations or external disruptions that may affect restaurant performance.

Marketing Recommendations

- Focus marketing efforts on high performing fast-food and beverage brands, as these restaurants consistently generated strong revenue and order volume.
- Promote highly rated restaurants through advertisements, food delivery applications, and loyalty campaigns to leverage customer trust and positive brand perception.
- Use cuisine popularity insights to develop targeted promotions around top performing categories such as North Indian, Chinese, South Indian, Biryani, Pizza, and Beverages.

Customer Experience Recommendations

- Encourage customers to leave reviews and ratings in order to improve restaurant visibility and strengthen customer trust.
- Continue prioritizing customer satisfaction initiatives, as the analysis suggested a positive correlation between higher ratings and increased revenue performance.

Business Strategy Recommendations

- Consider expanding partnerships or investment opportunities with restaurant brands and cuisine categories that consistently demonstrate strong customer demand and high revenue generation.
- Analyze the operational strategies used by top-performing restaurant chains to identify practices that could improve performance across lower performing restaurants.
- Review unusually high average order value transactions to identify potential data quality issues, reporting inconsistencies, or abnormal ordering behavior before using those values in strategic decision-making.